DOCUMIND AI

A RETRIEVAL AUGMENTED GENERATION SYSTEM FOR DOCUMENT QUESTION ANSWERING

SUBMITTED TO

SCHOOL OF COMPUTER SCIENCE ENGINEERING AND TECHNOLOGY, BENNETT UNIVERSITY

GREATER NOIDA, 201310, UTTAR PRADESH, INDIA

13th March 2026

DECLARATION

I HEREBY DECLARE THAT THE WORK WHICH IS BEING PRESENTED IN THE REPORT ENTITLED **"DOCUMIND AI: A RETRIEVAL AUGMENTED GENERATION SYSTEM FOR DOCUMENT QUESTION ANSWERING"**, IS AN AUTHENTIC RECORD OF OUR OWN WORK CARRIED OUT DURING THE PERIOD FROM JANUARY 2023 TO MARCH 2023 AT THE SCHOOL OF COMPUTER SCIENCE AND ENGINEERING AND TECHNOLOGY, BENNETT UNIVERSITY, GREATER NOIDA.

THE MATTERS AND RESULTS PRESENTED IN THIS REPORT HAVE NOT BEEN SUBMITTED BY US FOR THE AWARD OF ANY OTHER DEGREE ELSEWHERE.

.

SIGNATURE OF CANDIDATE

HARSH CHAUHAN

(E23CSEU2128)

ACKNOWLEDGEMENT

I WOULD LIKE TO TAKE THIS OPPORTUNITY TO EXPRESS OUR DEEPEST GRATITUDE TO PRASHANT SIR FOR GUIDING, SUPPORTING, AND HELPING US IN EVERY POSSIBLE WAY. WE WERE EXTREMELY FORTUNATE TO HAVE HIM AS OUR MENTOR AS HE PROVIDED INSIGHTFUL SOLUTIONS TO PROBLEMS FACED BY US THUS CONTRIBUTING IMMENSELY TOWARDS THE PROCESS OF THIS RESEARCH PAPER. WE WOULD ALSO LIKE TO EXPRESS OUR DEEPEST GRATITUDE TO VC, DEAN, HOD, FACULTY MEMBERS AND FRIENDS WHO HELPED US IN SUCCESSFUL COMPLETION OF THIS.

SIGNATURE OF CANDIDATE

HARSH CHAUHAN

(E23CSEU2128)

INTRODUCTION

Generative Artificial Intelligence has gained significant attention in recent years due to the development of powerful Large Language Models (LLMs).
These models are capable of generating human-like responses and answering complex questions.

However, traditional language models sometimes generate incorrect information because they rely only on their training data. Retrieval Augmented Generation (RAG)
is a technique that combines information retrieval with language models to improve accuracy.

This project presents DocuMind AI, a system that allows users to upload PDF documents and ask questions related to the document content.
The system retrieves relevant document information and generates accurate answers using a language model.

PROBLEM STATEMENT

Large documents such as research papers, notes, manuals, and reports contain significant amounts of information. Searching manually through these documents to find specific information can be time-consuming and inefficient.

Traditional chatbots cannot answer questions based on user-provided documents. Therefore, there is a need for a system that allows users to interact with documents and retrieve information efficiently using natural language queries.

OBJECTIVES

• Develop a Generative AI based document question answering system.
• Implement Retrieval Augmented Generation architecture.
• Allow users to upload PDF documents and ask questions.

LITERATURE REVIEW

Recent research in Natural Language Processing has focused heavily on transformer-based architectures. Vaswani et al. (2017) introduced the **Transformer model**, which became the foundation of modern language models. Lewis et al. (2020) introduced **Retrieval Augmented Generation (RAG)**, which integrates document retrieval with language models to improve response accuracy. Other studies have explored the use of **text embeddings** and **vector databases** for semantic search. Embedding models convert text into numerical vectors that capture semantic meaning.

Vector databases such as **FAISS** allow efficient similarity search across large embedding datasets, enabling rapid document retrieval.

SYSTEM ARCHITECTURE / METHODOLOGY

1. User uploads a PDF document.
2. The document text is extracted.
3. Text is split into smaller chunks.
4. Embeddings are generated for each chunk.
5. Embeddings are stored in a FAISS vector database.
6. User asks a question.
7. Relevant document chunks are retrieved.
8. The language model generates the final answer.

SYSTEM ARCHITECTURE

The architecture of the proposed system follows the RAG pipeline:

User → Upload PDF

↓

Document Processing

↓

Text Chunking

↓

Embedding Generation

↓

Vector Database (FAISS)

↓

User Query

↓

Similarity Search

↓

LLM Response Generation

TOOLS AND TECHNOLOGIES

Python

Streamlit

LangChain

FAISS Vector Database

Sentence Transformers

HuggingFace Transformers

IMPLEMENTATION

The system is implemented using Python and Streamlit for the user interface. The uploaded PDF is processed using a document loader and the text is divided into smaller chunks using a recursive text splitter.

Embeddings are generated using sentence transformer models and stored in a

FAISS vector database. When the user asks a question, the system retrieves relevant document chunks using similarity search and provides them to the language model to generate the final response.

RESULTS

The developed system successfully allows users to upload PDF documents and ask questions related to the content. The system retrieves relevant information from the document and generates context-aware answers.

This demonstrates the effectiveness of Retrieval Augmented Generation in improving the accuracy of language model responses.

CONCLUSION

DocuMind AI demonstrates how Generative AI can be used to build intelligent document-based question answering systems. By combining vector databases, embeddings, and language models, the system retrieves relevant information and generates accurate answers.

This project highlights the potential of Generative AI for building knowledge retrieval systems.

FUTURE SCOPE

- Support for multiple documents
- Integration with more powerful language models
- Deployment as a cloud-based web application
- Addition of chat history and memory

Advantages

- Reduces hallucination in AI responses
- Enables interaction with large documents
- Improves answer accuracy using document context
- Fast semantic search using vector databases

References

1. Vaswani, A. et al. (2017). Attention Is All You Need.
2. Lewis, P. et al. (2020). Retrieval Augmented Generation for Knowledge Intensive NLP Tasks.
3. HuggingFace Transformers Documentation.
4. LangChain Documentation.
5. FAISS Vector Database Documentation.